18 · How hearing happens

The words of this track, as it says them. Bold brown is the examiner.

How hearing occurs. Four marks.

Discuss how hearing occurs in humans, from the sound wave arriving at the ear, to the sound being perceived.

This one is a chain. Get the links in order, and the marks fall out.

The hook: collect, amplify, ripple, bend, fire, hear.

Collect. In the outer ear, the pinna collects sound waves and channels them along the external auditory canal to the tympanic membrane, which vibrates.

Amplify. In the middle ear, the vibration passes to the ossicles: malleus, then incus, then stapes. They amplify it onto the much smaller oval window.

Ripple. In the inner ear, the oval window vibrates the perilymph in the scala vestibuli and the scala tympani, and that passes the vibration on to the endolymph in the scala media, the cochlear duct.

Bend. The basilar membrane moves, and the hair cells of the organ of Corti bend against the tectorial membrane.

Fire. Bending the hairs is transduction. The hair cells are mechanoreceptors, and they turn the vibration into nerve impulses.

Hear. The impulses travel along the cochlear branch of the vestibulocochlear nerve, cranial nerve eight, to the auditory cortex in the temporal lobe, where the sound is heard.

The traps. It's the middle ear that amplifies: the ossicles. Her student-marked feedback slips and says the inner ear. Her deck says the middle ear, so write middle. Second: the hair cells sit on the basilar membrane, and bend against the tectorial membrane. Third: the auditory cortex is in the temporal lobe. The occipital lobe is vision.

Once more. How hearing happens.

The pinna collects sound down the canal, and the tympanic membrane vibrates. The ossicles, malleus, incus and stapes, amplify it onto the oval window. That moves the perilymph, then the endolymph, and the basilar membrane moves, so the hair cells of the organ of Corti bend against the tectorial membrane. That's transduction into nerve impulses, which travel along nerve eight to the auditory cortex in the temporal lobe.